

e-waste management by facilitating their evolution and integration in formal structures

The milestones achieved so far are:

1. Improved awareness:
 - Three WEEE Care! Initiative workshops in Bangalore supported by the Goethe Institute
 - National e-waste workshop in Delhi, hosted by MoEF
2. Improved stakeholder engagement:
 - Formation of the e-waste Agency (EWA) brings together industry, government and NGO to work on a sustainable e-waste management strategy for Bangalore
 - First national e-waste workshop held, defined a way forward
 - First national workshop on e-waste guidelines held, organised by MoEF
3. Improved estimates of e-waste:
 - Rapid assessments in Delhi and Bangalore of the quantities being generated, and identification of the e-waste recycling hot-spots
 - National-level desk study to assess e-waste quantities

A national-level assessment of electronics and electrical equipment waste (WEEE) by MoEF/CPCB/IRG/GTZ lists the top ten most polluting states and cities of India as shown in Tables III and IV. The figure are taken from the presentation of Dr Dilip B. Boralkar at *National Conference on E-Waste Management*, an Indo-German-Swiss E-Waste Initiative, at New Delhi on December 10, 2008.

The MAIT-GTZ study on e-waste found that 94 per cent of the organisations studied did not have any policy on disposal of obsolete IT products. Though many respondents (200 corporates and 400 households) were aware of e-waste, they were lacking in action.

Vinnie Mehta, executive director of the MAIT, in his presentation at *National Conference on E-Waste Management* (an Indo-German-Swiss E-Waste Initiative), listed the following legisla-

tions that cover different aspects of e-waste:

1. The hazardous waste (management and handling) rules, 1988 as amended in 2008 for toxic content—registration mandatory for recyclers
2. Municipal solid waste management and handling rules for non-toxic content
3. Basel convention for regulating trans-boundary movement
4. Foreign trade policy, which restricts import of second-hand computers and does not permit import of e-waste
5. Guidelines by Central Pollution Control Board (2008)

The guidelines notified in April 2008 identify and recognise:

1. Producers' responsibility
2. RoHS (restriction on hazardous substances)
3. Best practices
4. Insight into technologies for various levels of recycling

Mehta said that the guidelines explicitly mention the need for a separate legislation for implementing producers' responsibility. He said that e-waste is 'distinct' as it is an end-of-consumption waste while hazardous waste results from a distinct industrial process. The Environment Protection Act provides for separate regulations for waste with 'distinct' characteristics—Biomedical Wastes (M&H) Rules 1998, Batteries (M&H) Rules 2001, etc.

Advocating a separate legislation for e-waste, he said that in his recent presentation to members of the parliament he has emphasised that e-waste value chain is rather complex as it involves multiple players—producers, distributors, retailers, end consumers, collection system and recyclers—while hazardous waste chain involves only the occupier/generator and the operator. Recovery of non-ferrous metals and reprocessing of used oil are the only two major activities in hazardous waste recycling, while e-waste recycling involves refurbishment for reuse, dismantling and precious



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